

AMIT TOMAR

Generalist engineer with 13+ years of experience in UI, 3D Computer Graphics and Backend

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🐙 github.com/Amit-Tomar

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📖 stackoverflow.com/u/1093223

EXPERIENCE

Principal Engineer

Autodesk

📅 May 2025 – Present

📍 Bangalore

- Autodesk Operations Solutions: Digital Factory twins and simulations.

Software Development Engineer (SDE-III)

Livspace

📅 Nov 2020 – May 2025

📍 Bangalore

- Built highly scalable REST APIs for ray traced offline rendering of 3D assets using Blender, in cloud.
- Contributed features to internal home interiors design configurator, which helps designers with floor planning, room design and high quality 3D product visualization, in real time, on web.

Software Engineer

CK-12 Foundation

📅 Aug 2019 – Oct 2020

📍 Bangalore

- Developed tools and highly interactive 2D/3D content for K-12 education.

Group Development Manager

Avataar.ai

📅 July 2015 – July 2019

📍 Bangalore

Part of the initial 5 members tech team at 3D fashion and AR/VR startup. Got hands dirty in all aspects of product and software development, acting as conduit between tech, design and 3D teams.

- Wrote RESTful APIs to manage user Avataars and 3D inventory.
- Setup 3D assets generation pipeline, providing storage, optimization and retrieval of high quality, low poly virtual 3D garments & Avataars.
- ThreeJS based configurator for 3D Avataars and apparels.
- Performance analysis and enhancement of internal cloth simulation engine.

Systems Engineer

Tata Consultancy Services

📅 Dec 2010 – July 2013

📍 Embedded Innovation Lab, Bangalore

Part of the capability demonstrator team at Innovation Lab, creating proof of concepts by rapid prototyping for prospective customers.

- Implemented HMI for various applications like Media player, car controls, digital instrument cluster etc. for in house In-vehicle Infotainment system.
- Involved in design & development of underlying technology agnostic middleware frameworks for inter process communication & HMI.

SKILLS



Frontend

Javascript, D3.js, Processing.js, SVG



Computer Graphics

WebGL, Three.js, Blender scripting, OpenGL



Backend

Python-Flask, RESTful APIs.



Systems

C++, Qt Framework, QML



Cloud

AWS, Boto3

ACHIEVEMENTS



Patent

- Aluru, S, G Baid, and A Tomar (2020). "Virtual representation creation of user for fit and style of apparel and accessories." URL: <https://patents.google.com/patent/US10777020B2/en>.



Publication

- Agarwal S. Tomar A., Sreevalsan-Nair J (2017). "NodeTriX-Multiplex: Visual Analytics of Multiplex Small World Networks". In: *Complex Networks & Their Applications V*, Springer, Cham. URL: <https://bit.ly/3TnqYRA>.



Others

- 99.08 Percentile (GATE CSE 2013).
- Contributed around 1000 LOC to open source GNU package GCompris, porting two games to Qt Framework.

EDUCATION

M.Tech. in IT

IIIT Bangalore

📅 2013 – 2015

B.Tech. (Honors) in CSE

Meerut Institute of Engineering & Technology

📅 2006 – 2010